

COMPUTATIONAL GENOMICS FINAL PROJECT

Who left this DNA?

A mystery Nanopore fecal sample, six possible host animals, one microbial co-star, and a small pile of evidence.

Group D

Host origin, microbiome, assembly, and validation from curated project outputs.

THE BRIEF

A final project with detective work built in

Given:

Long-read sequencing data from an unknown fecal sample.

Needed:

A defensible host call, microbial profile, assembly attempt, and reproducible notebook.

6

candidate host genomes

300k

reads used for the first host screen

1

dominant microbial species

1

reproducible notebook pipeline

ANALYSIS MAP

From messy reads to a biological story

1

Check the reads

Are they long enough and clean enough to trust?

2

Find the host

Align reads to the six candidate genomes.

3

Inspect a locus

Use an IGV-style view to look at reads, not just totals.

4

Classify microbes

Ask what else appears in the fecal DNA.

5

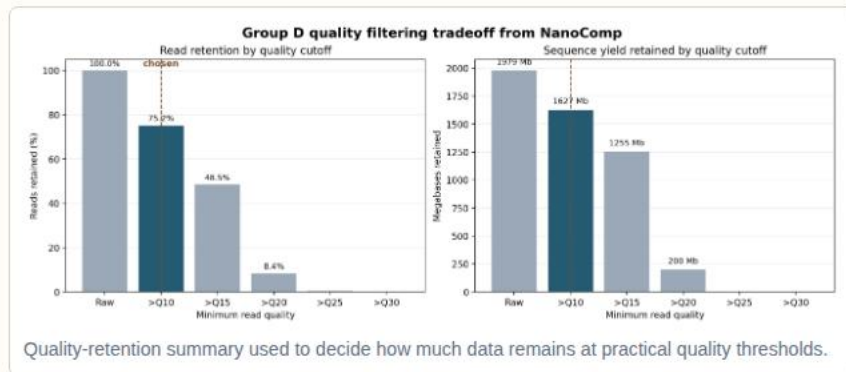
Assemble and validate

Reconstruct the dominant microbe and compare it to a reference.

The goal was not to make a fancy tool zoo. It was to make each step answer the next practical question.

QUALITY CONTROL

First question: are the reads usable?



Answer:

Yes. The reads are noisy, as Nanopore reads usually are, but still usable for long-read alignment and microbial screening.

Why these reports?

MinKNOW describes the sequencing run itself. NanoComp and NanoPlot summarize the reads after the fact, which is better for reproducible QC in the notebook. FastQC is common for short reads, but less informative for Nanopore length distributions.

QC RECEIPTS

The reports stayed visible, not hidden in a folder

MinION Mk1B MN41615 Sequencing final run report  Oxford Nanopore Technologies

May 28, 26, 10:22 PM UTC+3:00 — May 31, 26, 10:25 PM UTC+3:00 · Group 0 · no_sample_id · MN41615
Protocol run ID: 06810787-B06d-4d18-854b-c5692c973cdc

[Run summary](#) | [Run configuration](#) | [Sequence output](#) | [Run health](#) | [Run log](#)

Run summary

DATA OUTPUT	BASECALLING	RUN DURATION
Estimated bases	Reads called	Run time

MinKNOW: run-level sequencing summary.

Summary Statistics | Plots | [Report issue on Github](#)

NanoComp report

Summary statistics

Feature	group_a	group_c
General summary		
Mean read length	1,832.1	1,837.5
Mean read quality	13.6	10.6

NanoComp: side-by-side read quality summaries.

Summary Statistics prior to filtering | Summary Statistics after filtering | Plots | [Report issue on Github](#)

NanoPlot reports

Summary statistics prior to filtering

Metrics	dataset

NanoPlot: read length and quality distributions.

This is the boring slide that prevents exciting wrong conclusions later.

SUSPECT LINEUP

Six animals walked into the alignment step



House mouse

Good cameo, weak evidence.



Kangaroo

Memorable, but not our sample.



Domestic cat

Quietly sitting on the strongest signal.



Norway rat

Similar enough to test, not enough to win.



Golden jackal

Second place in some counts.



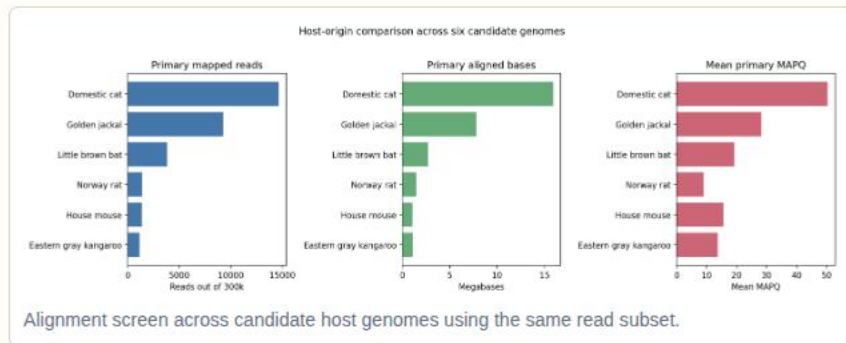
Little brown bat

Useful negative control energy.

The mouse got a slide. The cat got the evidence.

FIRST TRY

Let the reads vote: which genome do they align to?



Domestic cat had the strongest host signal.

It had the most primary mapped reads and the most aligned bases among the candidate genomes.

Why minimap2?

It is fast, widely used, and tuned for long noisy reads. BWA-MEM is excellent for many Illumina-style tasks, but minimap2 is the practical default for Nanopore host screening. BLAST would be too slow at this scale.

SECOND TRY

Not just more reads. Better reads.

Candidate	Primary mapped	Aligned bases	Mean MAPQ
1 Domestic cat	14,669	15.9 Mbp	50.3
2 Golden jackal	9,251	7.8 Mbp	28.3
3 Little brown bat	3,818	2.7 Mbp	19.2
Mouse / rat / kangaroo	1,155-1,393	1.1-1.5 Mbp	9.0-15.6

MAPQ is the confidence check.

A higher MAPQ means the aligner is more confident that a read belongs at that location.

Cat wins twice.

More mapped signal and much higher confidence than the nearest alternative.

HOST REVEAL

The sample origin is domestic cat

The candidate lineup was fun. The alignment scorecard made the call.

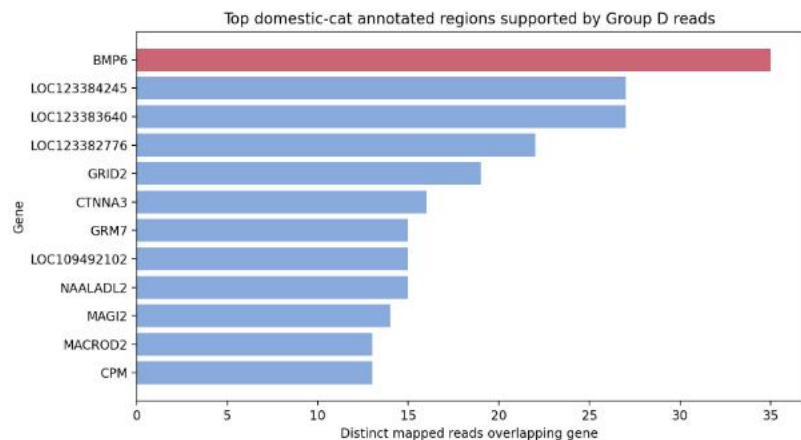
Top mapped reads

Top aligned bases

Top MAPQ



Where did the cat reads land?



BMP6 had the highest read count among annotated cat gene overlaps.

Top locus: BMP6

35 reads overlapped BMP6, with high best MAPQ.

Important caveat:

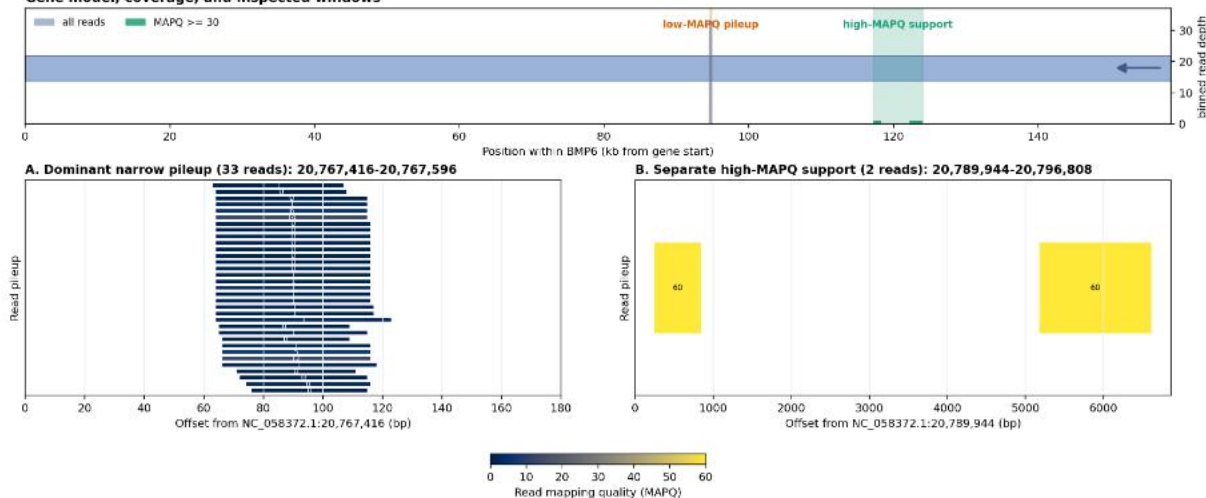
This is not a claim about BMP6 biology in the animal. It says the strongest annotated host overlap in this read set was near BMP6.

The slide where we look at the reads, not only the bar chart

BMP6 host-region evidence: genome-browser-style view

NC_058372.1:20,672,770-20,831,131 | 158.4 kb gene | 35 overlapping reads | 2 reads with MAPQ ≥ 30

Gene model, coverage, and inspected windows



IGV-style BMP6 locus panel. This provides a visual sanity check of read placement over the locus.

Why an IGV-style panel?

Summary tables can hide awkward alignments. A genome-browser view lets us see whether the reads form a plausible local signal.

Read it carefully:

The signal is concentrated and should be treated as supporting evidence, not as a functional discovery.

A fecal sample rarely travels alone

After host screening, the next question was: who else is in there?

For that, we used taxonomic classification rather than host-style whole-genome alignment.

Why Kraken2?

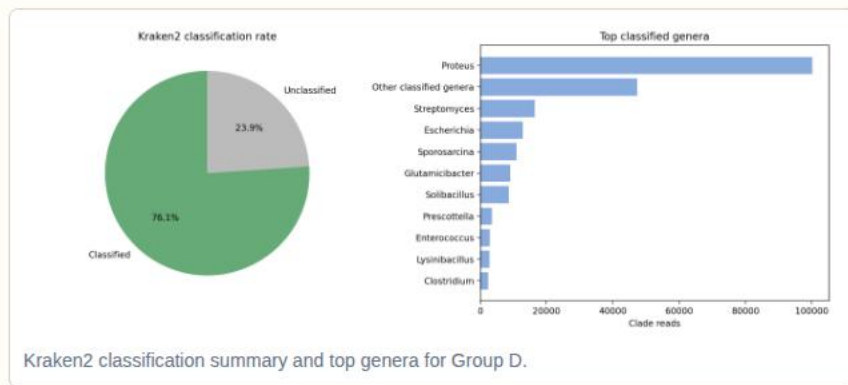
Kraken2 is fast and works well when the question is "what taxa are represented here?" Centrifuge is another lightweight classifier, MetaPhlAn is strong for marker-gene profiles, and KrakenUniq adds extra support information. Kraken2 was the best fit for a quick, reproducible species screen.

Plain version:

Host alignment asks "which animal genome?" Kraken2 asks "which microbial names do these reads most resemble?"

MICROBIAL SCORECARD

Proteus dominates the genus-level profile



35.11%

Proteus at genus level

5.77%

Streptomyces

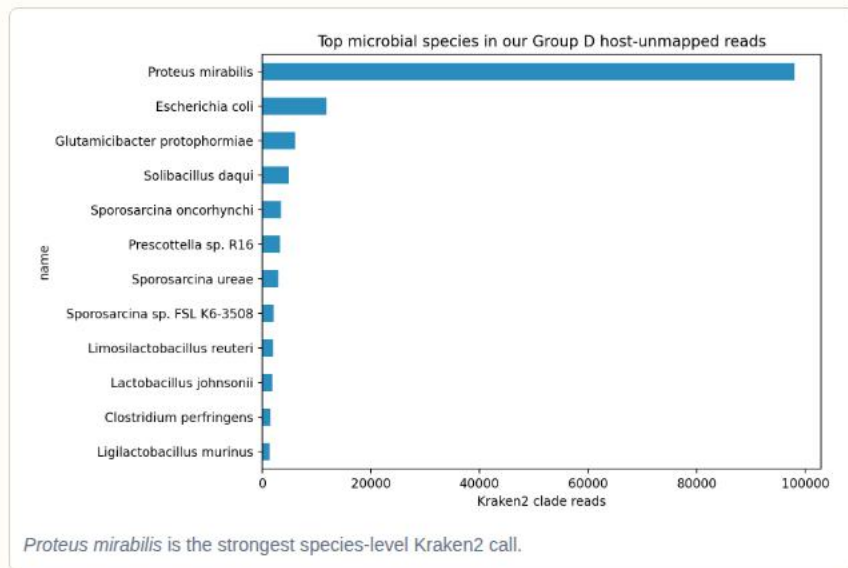
4.49%

Escherichia

3.84%

Sporosarcina

The dominant species is *Proteus mirabilis*



Top species call:

Proteus mirabilis, 34.35% of reads in the Kraken2 report.

The biological story so far:

A domestic cat fecal sample with a strong *Proteus mirabilis* microbial signal.

Can we reconstruct the microbial co-star?

Why assemble *Proteus mirabilis*?

It had the strongest species-level signal, so it was the best candidate for a focused assembly attempt.

Why Flye?

Flye is built for noisy long reads and is commonly used for bacterial Nanopore assemblies. Canu is powerful but heavier, Raven is fast and lightweight, and Shasta is excellent in some long-read contexts. Flye was a balanced choice for this final-project dataset.

3.86 Mbp

assembly size

24

contigs

577,987 bp

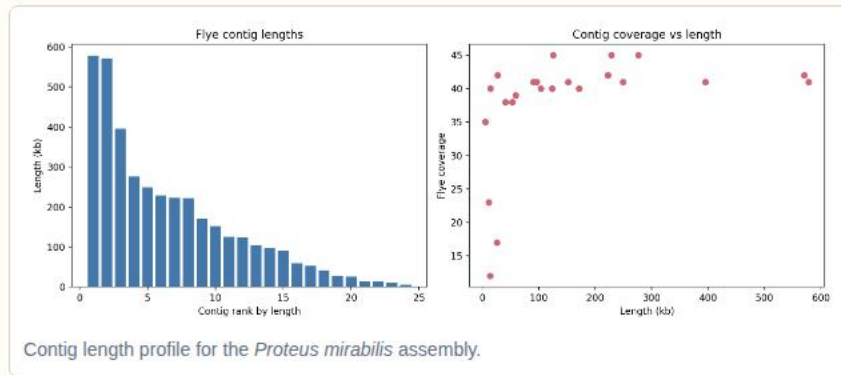
longest contig

249,377 bp

N50

ASSEMBLY RESULT

The assembly is fragmented, but biologically useful



What this means:

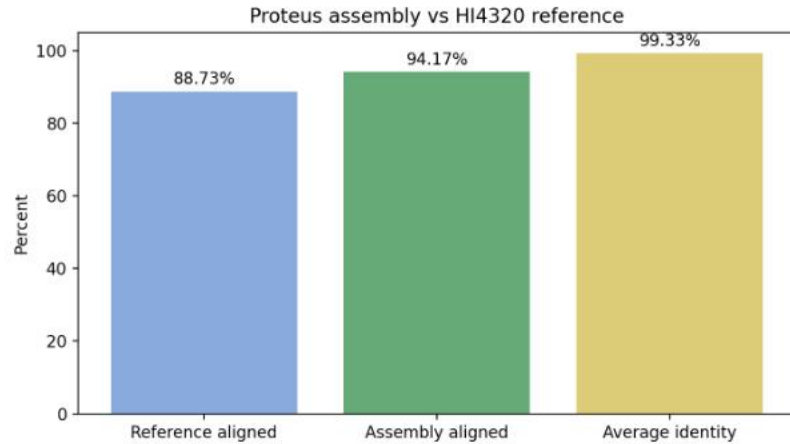
A perfect bacterial assembly would be closer to one complete chromosome. Here we got a partial but coherent assembly from a mixed fecal sample.

Good enough for:

Checking whether the dominant microbe can be reconstructed and compared against a reference.

VALIDATION

Does the assembly look like *Proteus mirabilis*?



dnadiff comparison between the assembly and a *Proteus mirabilis* reference.

Average identity: 99.33%

The assembled sequence aligns very closely to the reference where it aligns.

Coverage is not complete.

88.73% of the reference and 94.17% of the assembly aligned in the comparison.

Why dnadiff?

It gives a compact reference-vs-assembly comparison. QUAST is a strong alternative for broader assembly QC, and minimap2 plus custom summaries would also work, but dnadiff gives the most direct identity summary here.

WHAT WE TRUST

Strong conclusions, careful caveats

We trust these results

- Host origin is domestic cat based on read count, aligned bases, and MAPQ.
- *Proteus mirabilis* is the dominant species-level microbial call.
- The Proteus assembly is close to a reference over aligned regions.

We are careful about these points

- BMP6 is a read-overlap observation, not a functional phenotype claim.
- Kraken2 depends on the database and can misclassify reads from close relatives.
- The assembly is fragmented and from a mixed sample, so it is not a finished genome.

FINAL ANSWER

The short version

Host:

Domestic cat.

Top host locus:

BMP6, with a cautious interpretation.

Dominant microbe:

Proteus mirabilis.

Assembly:

3.86 Mbp, 24 contigs, N50 249,377 bp,
99.33% identity.

The cat did it.

Proteus mirabilis left the microbial fingerprints.

Full report

Curated GitHub Pages report

Notebook

Rendered reproducible analysis

PDF deck

Downloadable presentation